PianoTest — piano di test MuonScope sul bus MIKILAB (REV5.0). Vedi README.md.

Rivelatore

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PREAMPLIFICATORE DI CARICA (OPA4172 A, Cf 1 pF + Rf 100M di default) -> shaper CR-RC tau 1 us: C invertente x10 (CR), D non invertente x5 (RC) = x50. Base 1V25 bufferizzata da B.

BPW34 a 30-36 V: Cd ~8-9 pF, MIP ~1-1.3 fC -> ~1 mV su CSA_OUT (negativo), ~50 mV positivi su SHAPER_OUT sopra 1.25 V. Filtro ADC 47R + 1 nF.

OPZIONE JFET (DNP): BF862 a source comune + BF862 cascode. Diretto = R120/R121 montate; JFET = togliere R120/R121, montare R122/R123, Q101/Q102 e la polarizzazione (valori da tarare).

U101
OPA4172ID
OPA_POS
OPA_NEG
1V25_REF
SH_BASE
SH_SUM
SH_RC
SH_FB
+5V
GND
+INA
-INA
+INB
-INB
+INC
-INC
+IND
-IND
V+
V-
OUTA
OUTB
OUTC
OUTD
CSA_OUT
SH_BASE
SH_C
SHAPER_OUT
D101
VBPW34S
CSA_IN
A
K
DIODE_K
U102
ADS8326IDGKT
2V5_REF
ADC_IN
GND
+3V3
REF
+IN
-IN
VDD
GND
CS
DOUT
DCLOCK
CS_ADC
MISO_L
SCLK_L
U103
TMP117AIDRVR
V+
GND
ADD0
EP
SCL
SDA
ALERT
I2C_SCL
I2C_SDA
PL_FAULT
Q101
BF862
JFET_C
JFET_S
Q102
BF862
CASC_B
JFET_D
JFET_C
VBIAS_SH
R101
1k
VBIAS_DIODE
JFET_REF
R123
0R
OPA_NEG
JFET_S
R124
750R
GND
JFET_REF
R129
15k
GND
CSA_OUT
C102
1nF C0G
SH_CD
SH_FB
R113
4.02k
SHAPER_OUT
CSA_OUT
R107
100R
SMA_PRE
2V5_REF
C111
47uF
GND
VBIAS_DIODE
C101
100nF/100V C0G
GND
JFET_S
C124
10uF
GND
JFET_REF
C129
1uF
GND
SH_CD
R103
1k
SH_SUM
SHAPER_OUT
R106
47R
ADC_IN
J101
SMA preampl.
SMA_PRE
1
2
GND
+5V
C112
1uF
GND
SHAPER_OUT
R108
100R
SMA_SH
VBIAS_DIODE
R102
1M
DIODE_K
JFET_D
R125
1k
+5V
SH_SUM
R104
10k
SH_C
ADC_IN
C107
1nF C0G
GND
J102
SMA shaper
SMA_SH
1
2
GND
+5V
C113
100nF
GND
DIODE_K
C103
100nF/100V C0G
GND
CASC_B
R126
10k
+5V
SH_SUM
C105
100pF C0G
SH_C
TP101
CSA_OUT
TP102
SHAPER_OUT
TP103
VBIAS_DIODE
CSA_OUT
SHAPER_OUT
VBIAS_DIODE
+3V3
C114
1uF
GND
CSA_IN
R120
0R
OPA_NEG
CASC_B
R127
4.3k
GND
SH_C
R105
1k
SH_RC
+3V3
C115
100nF
GND
1V25_REF
R121
0R
OPA_POS
CASC_B
C127
1uF
GND
SH_RC
C106
1nF C0G
SH_BASE
+3V3
C116
100nF
GND
JFET_D
R122
0R
OPA_POS
+5V
R128
10k
JFET_REF
SH_FB
R112
1k
SH_BASE
SH_BASE
C117
100nF
GND

SMA con 100R in serie: oscilloscopio a 1 MOhm. C101/C103/C506 C0G 100 nF 100 V 1206 (TDK C3216C0G2A104J160AE): niente X7R, perdono nA a 30 V.

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RETROAZIONE: R201 100M sempre presente (DC definita anche con MCP in reset). Rf 10k..10M con TMUX1112 (chiuso con SEL=1, pull-down 100k = aperti al reset).

Cf: C201 1 pF fisso (default) + banchi BGSA142: -IN -> Cf -> RFx -> RFC -> 10 nF C0G -> uscita. 100M verso GND su ogni RFx e RFC. CTL tramite partitore 10k/22k (3.3 V -> 2.27 V, pull-down al reset).

U201
MCP23S17-E/ML
+3V3
GND
GND
CS_MCP
SCLK_L
MOSI_L
MISO_L
GND
GND
GND
MCP_RST
MCP_INTA
VDD
VSS
EP
CS
SCK
SI
SO
A0
A1
A2
RESET
INTB
INTA
GPA0
GPA1
GPA2
GPA3
GPA4
GPA5
GPA6
GPA7
GPB0
GPB1
GPB2
GPB3
GPB4
GPB5
GPB6
GPB7
PL_FAULT
TRIG_MCP
SEL_R1
SEL_R2
SEL_R3
SEL_R4
SEL_R5
SEL_R6
SEL_R7
SEL_R8

U202
TMUX1112RSVR
+5V
GND
SEL_R1
SEL_R2
SEL_R3
SEL_R4
VDD
GND
SEL1
SEL2
SEL3
SEL4
NC
NC
S1
D1
S2
D2
S3
D3
S4
D4
CSA_IN
RF_1
CSA_IN
RF_2
CSA_IN
RF_3
CSA_IN
RF_4

U203
TMUX1112RSVR
+5V
GND
SEL_R5
SEL_R6
SEL_R7
SEL_R8
VDD
GND
SEL1
SEL2
SEL3
SEL4
NC
NC
S1
D1
S2
D2
S3
D3
S4
D4
CSA_IN
RF_5
CSA_IN
RF_6
CSA_IN
RF_7
CSA_IN
RF_8

U204
BGSA142GN12
+3V3
GND
GND
CTL0
CTLA1
CTLA2
VDD
GND
GND
CTL0
CTL1
CTL2
RFC1
RFC2
RF1
RF2
RF3
RF4
CFA_COM
CFA_COM
CFA1
CFA2
CFA3
CFA4

U205
BGSA142GN12
+3V3
GND
GND
CTL0
CTLB1
CTLB2
VDD
GND
GND
CTL0
CTL1
CTL2
RFC1
RFC2
RF1
RF2
RF3
RF4
CFB_COM
CFB_COM
CFB1
CFB2
CFB3
CFB4

CSA_IN
R201
100M
CSA_OUT

RF_6
R216
2.2M
CSA_OUT

SEL_R1
R221
100k
GND

SEL_R8
R228
100k
GND

CSA_IN
C231
0.5pF C0G
CFA1

CFA4
R234
100M
GND

CSA_IN
C241
10pF C0G
CFB1

CFB4
R244
100M
GND

CSA_IN
C201
1pF C0G
CSA_OUT

RF_7
R217
4.7M
CSA_OUT

SEL_R2
R222
100k
GND

CFA1
R231
100M
GND

CFA_COM
C236
10nF C0G
CSA_OUT

CFB1
R241
100M
GND

CFB_COM
C246
10nF C0G
CSA_OUT

RF_1
R211
10k
CSA_OUT

RF_8
R218
10M
CSA_OUT

SEL_R3
R223
100k
GND

CSA_IN
C232
1pF C0G
CFA2

CFA_COM
R236
100M
GND

CSA_IN
C242
22pF C0G
CFB2

CFB_COM
R246
100M
GND

RF_2
R212
47k
CSA_OUT

SEL_R4
R224
100k
GND

CFA2
R232
100M
GND

CFB2
R242
100M
GND

RF_3
R213
100k
CSA_OUT

SEL_R5
R225
100k
GND

CSA_IN
C233
2.2pF C0G
CFA3

CSA_IN
C243
47pF C0G
CFB3

RF_4
R214
470k
CSA_OUT

SEL_R6
R226
100k
GND

CFA3
R233
100M
GND

CFB3
R243
100M
GND

RF_5
R215
1M
CSA_OUT

SEL_R7
R227
100k
GND

CSA_IN
C234
4.7pF C0G
CFA4

CSA_IN
C244
100pF C0G
CFB4

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EMETTITORE: LTC6993-2 (durata con AD5245 100k reostato + 47k) -> BSS138 -> LED IR alimentato da +5V (non piu da V_BIAS).

TRIG da MCP GPA7 (tramite R304) o dal connettore J301. AD5245 su I2C del bus, indirizzo 0x2C (AD0 a GND).

U301
LTC6993CS6-2
+3V3
GND
TRIG_INPULSE
LTC_SET
GND
V+
GND
TRIG
SET
DIV
OUT
LTC_OUT

U302
AD5245BRJZ100
+3V3
GND
I2C_SCL
I2C_SDA
GND
VDD
GND
SCL
SDA
AD0
A
W
B
RHEO_W
GND

Q301
BSS138
Q_G
G
D
S
LED_K
GND

D301
VSMY2940G
LED_A
A
C
LED_K

RHEO_W
R301
47k 1%
LTC_SET

LTC_OUT
R302
47R
Q_G

Q_G
R303
10k
GND

+5V
R305
220R
LED_A

TRIG_MCP
R304
1k
TRIG_INPULSE

TRIG_INPULSE
R306
10k
GND

J301
TRIG est.
TRIG_INPULSE
GND

+3V3
C301
100nF
GND

+3V3
C302
100nF
GND

+5V
C303
10uF
GND

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BIAS LOCALE: DAC61402 alimentato da VHV_36V (bus pin 8) con RC 100R + 2x1uF/100V. OUTA -> shunt 10k (INA239) -> VBIAS_SH -> RC 1k/100nF vicino al diodo.

INA239 con shunt 10k: ADCRANGE=1 -> 1.25 uV/LSB = 125 pA/LSB, fondo scala 4.1 uA; ADCRANGE=0 -> 500 pA/LSB, 16.4 uA. Errori: Vos 5 uV max (0.5 nA), IB 0.1 nA tip / 2.5 nA max.

DAC: AVDD -36 V -> uscita max -34.5 V (headroom 1.5 V). OUTB su VBIAS_B (riserva / secondo diodo). CCOMP DNP.

U501
DAC61402RHBR
AVDD_DAC
+5V
+3V3
DAC_REF
GND
GND
GND
GND
GND
CS_DAC
SCLK_L
MOSI_L
MISO_L
GND
+3V3
+3V3
PL_FAULT
AVDD
DVDD
IOVDD
REFIO
REFGND
AVSS
AGND
GND
EP
SYNC
SCLK
SDIN
SDO
LDAC
CLR
RST
FAULT
OUTA
SENSEPA
SENSENA
CCOMPA
OUTB
SENSEPB
SENSENB
CCOMPB
NC
NC
NC
NC
NC
NC
NC
NC
DAC_OUTA
DAC_OUTA
GND
CCOMPA
DAC_OUTB
GND
CCOMPB
U502
INA239AIDGSR
+3V3
GND
CS_INA
SCLK_L
MOSI_L
MISO_L
PL_FAULT
VS
GND
CS
SCLK
MOSI
MISO
ALERT
IN+
IN−
VBUS
DAC_OUTA
VBIAS_SH
VBIAS_SH
VHV_36V
R501
100R
AVDD_DAC
DAC_OUTA
R502
10k 0.1%
VBIAS_SH
+5V
C507
100nF
GND
AVDD_DAC
C501
1uF/100V
GND
DAC_OUTB
R503
1k
VBIAS_B
+3V3
C508
100nF
GND
AVDD_DAC
C502
1uF/100V
GND
VBIAS_B
C506
100nF/100V C0G
GND
DAC_REF
C509
100nF
GND
AVDD_DAC
C503
100nF/100V
GND
+3V3
R504
10k
PL_FAULT
+3V3
C510
100nF
GND
CCOMPA
C504
470pF/100V
DAC_OUTA
CCOMPB
C505
470pF/100V
DAC_OUTB
TP501
DAC_OUTA
TP502
VBIAS_B
DAC_OUTA
VBIAS_B

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BUS MIKILAB REV5.0 – MAIN 28 pin. J_IN dal piano precedente. J_OUT verso il successivo: tutto passante tranne ID (25–27), che esce come ID+1 (74HC283).

Piano selezionato se PREFIX = ID_IN (74HC85 QA=B → E3 del 138). CS_MASTER su –E1, –E2 a GND. Y0 ADC, Y1 DAC, Y2 INA, Y3–Y5 test point, Y6 EEPROM ID, Y7 MCP23S17.

MISO verso il bus tramite 74LVC1G125, abilitato solo con piano selezionato e CS_MASTER basso. ALERT# open-drain (74LVC1G07) da INTA (MCP con MIRROR=1). Pull-up I2C e ALERT# sulla scheda master.

FPC: XUNPU FPC-05F-28PH20 (LCSC C2856807). footprint PianoTest:FPC-05F-28PH20 importato da EasyEDA. Contatti in basso.

J401
J_IN MAIN

J402
J_OUT MAIN

U401
74HC283

U402
74HC85

U403
74HC138

U404
SN74LVC2G17DBVR

U405
SN74LVC1G125DBVR

U406
SN74LVC1G332DBVR

U407
SN74LVC1G07DBVR

U408
25AA02E48T-I/SN

TP401
CS_Y3

TP402
CS_Y4

TP403
CS_Y5

R401
0R

R402
100k

R403
100k

R404
100k

R405
100k

R406
100k

C401
100nF

C402
100nF

C403
100nF

C404
100nF

C405
100nF

C406
100nF

C407
100nF

C408
100nF

C409
10uF

C410
10uF

C411
100nF/100V

C412
100nF/100V

C413
1uF

C414
1uF

D401
D402
D403
D404
D405
D406
D407
D408
TPDLE10B06

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